AIM:

To write a Python program to implement Depth First Search (DFS) traversal.

ALGORITHM :

1. Start the program.
2. Create a graph using a dictionary.
3. Initialize an empty set to store visited vertices.
4. Visit the starting vertex and print it.
5. Mark the vertex as visited.
6. Recursively visit all unvisited neighbouring vertices.
7. Repeat until all reachable vertices are visited.
8. Stop the program.

PROGRAM :

File Edit Format Run Options Window Help

```
def dfs(graph, start, visited=None):
    if visited is None:
        visited = set()

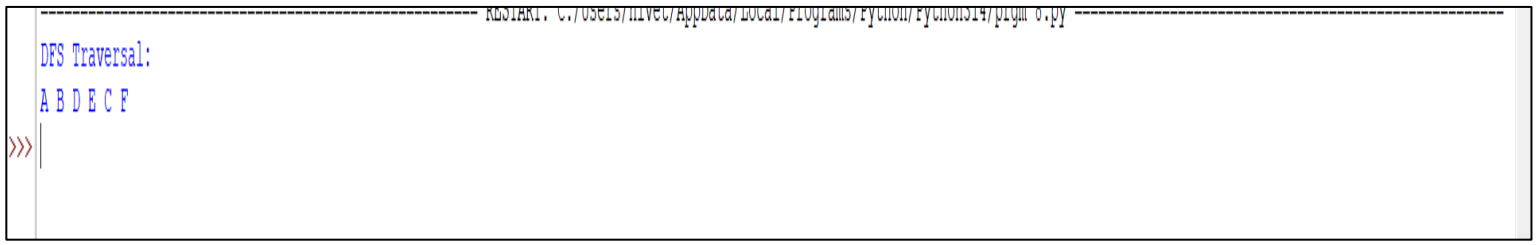
    if start not in visited:
        print(start, end=" ")
        visited.add(start)

        for neighbour in graph[start]:
            dfs(graph, neighbour, visited)

# Graph representation
graph = {
    'A': ['B', 'C'],
    'B': ['D', 'E'],
    'C': ['F'],
    'D': [],
    'E': [],
    'F': []
}

print("DFS Traversal:")
dfs(graph, 'A')
```

OUTPUT :



RESULT :

Thus, the Python program to implement DFS traversal was executed successfully.